

RGMII clocking application note for GEM GXL

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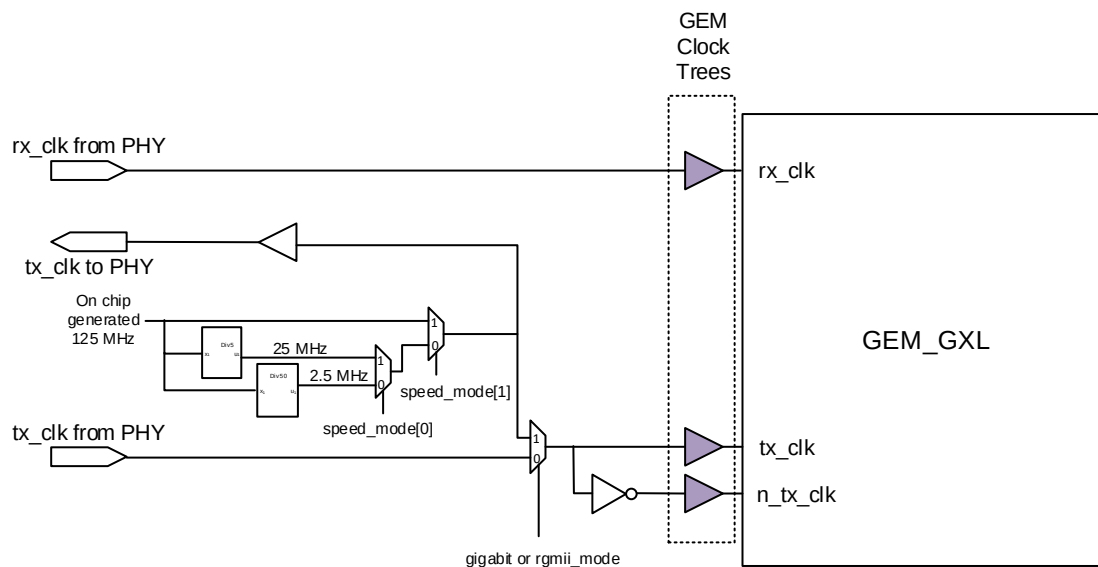
RGMII clocking

The RGMII standard specifies a source synchronous clock. In the egress direction, the data and clock are transmitted from the SoC chip containing the GEM GXL to the RGMII PHY chip. In the ingress direction, the data and clock are transmitted from the RGMII PHY chip to the SoC chip containing the GEM GXL. The data is sampled at the receiving end using the same edge of the clock on which it was generated. To do this safely, the clock must have a longer path delay than the data. The RGMII standard supports two different versions. Version 1.3 relies on a PCB trace delay of 1.5 to 2.0 ns for the clock, whereas version 2.0 and later offers the option of introducing the delay on-chip at the source. Devices which support the internal delay (ID) are referred to as RGMII-ID. The GEM GXL does not have any clock generation logic within the IP and therefore it is the responsibility of the implementer to drive the `tx_clk` transmit clock output from the SoC appropriately.

Most RGMII PHY chips will provide support for delaying the input clock and output clock. If this is available, the integrator may decide to not implement any egress clock delays (either inside the SoC, or on the PCB trace). The GEM GXL IP supports both versions of the RGMII spec but it is the responsibility of the implementer to insert the appropriate delay to support RGMII-ID if it is needed. A buffer with programmable delay could for example be used to drive the delayed `tx_clk` signal going to the RGMII PHY.

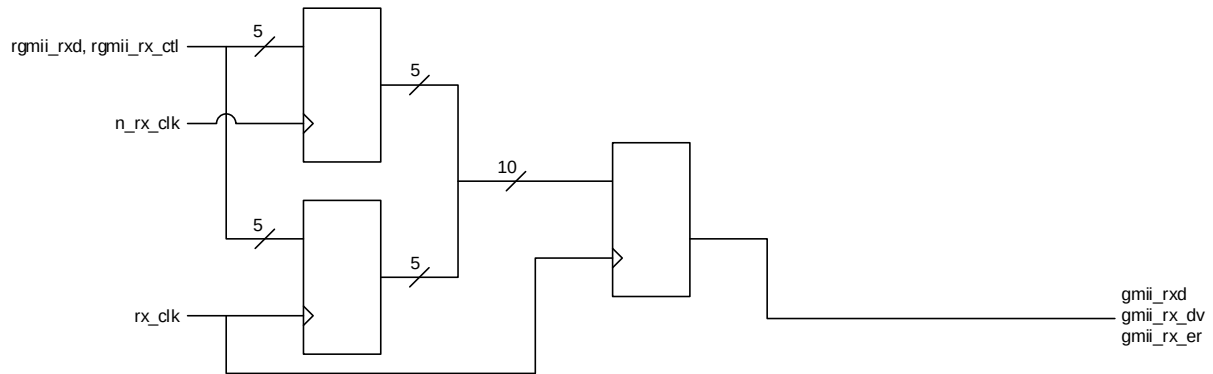
The other main difference between version 1.3 and 2.0 is the type of pads used. Version 2.0 utilizes 1.5 V HSTL pads, whereas version 1.3 uses 2.5 V CMOS pads.

An example clocking diagram is shown below. In RGMII or gigabit mode, `tx_clk` is source synchronous and needs to be output from the ASIC along with the transmit data signals. In 10/100 MII mode, `tx_clk` is sourced externally from the PHY.



RGMII Receive

It is the responsibility of the system clock controller to provide the `rx_clk` source and the inverted `n_rx_clk` source to the IP. The following diagram shows the demultiplexing of the RGMII DDR receive signals.

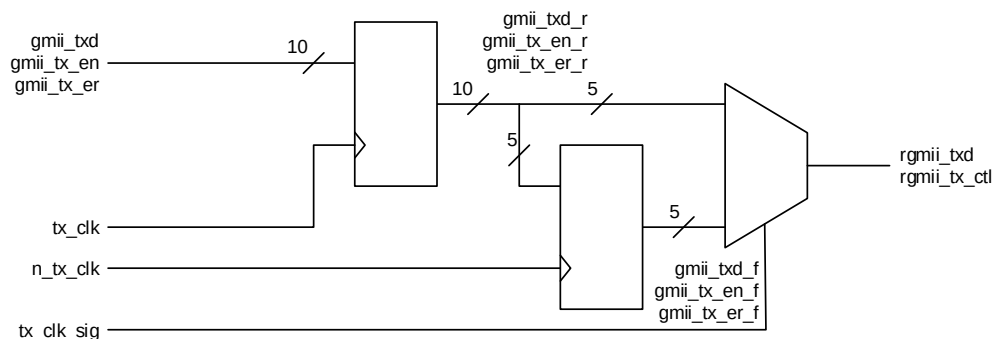


It can be assumed that at the receiving end within the SoC containing the GEM GXL IP, the clock from the RGMII PHY will already be delayed with respect to the data (centre-aligned). From the RGMII standard, this delay must be at minimum 1ns at the receiving end. There is also a hold requirement of at least 1ns, so this infers the delay between clock and data will always be between 1ns and 3ns. We assume that differences between the clock/data trace delay, pad delays, pin capacitance, and rise/fall time are negligible. Within the SoC, we must ensure that the delays from `rgmii_rxd/rgmii_rx_ctl` input and the generated `rx_clk/n_rx_clk` clocks to the first flops are constrained appropriately to ensure the data can be sampled safely – a difference of more than 1ns will cause a problem.

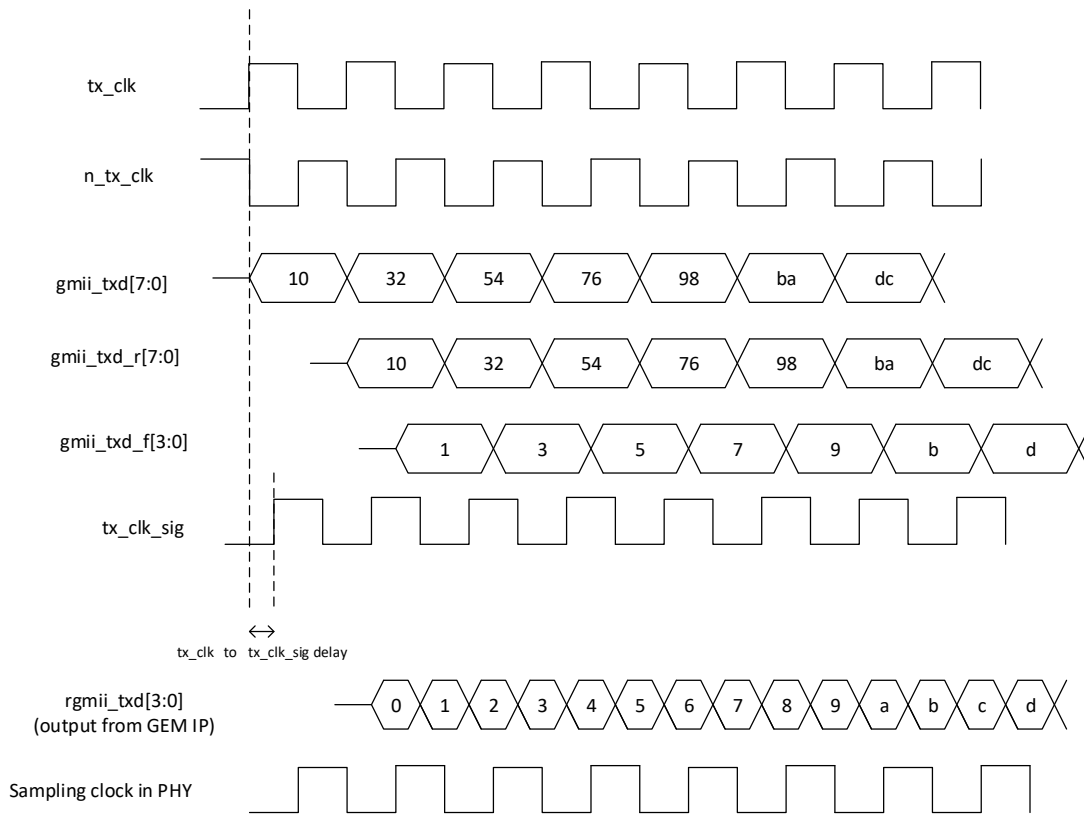
RGMII Transmit

For RGMII operation, the system clock controller is required to provide the `tx_clk` source and the inverted `n_tx_clk` source to the IP, and the external `tx_clk` output for the PHY. To ensure that the skew requirements of the RGMII are met, for RGMII version 1.3 the delays on this external clock output should be matched to the RGMII transmit data and control outputs (edge-aligned), for RGMII-ID the transmit clock output should be delayed 2ns relative to these outputs (centre-aligned).

The `rgmii_txd` and `rgmii_tx_ctl` outputs to the RGMII PHY are DDR in nature. The GEM_GXL implements this using two flip-flops and a multiplexor to drive the final pad outputs as follows:



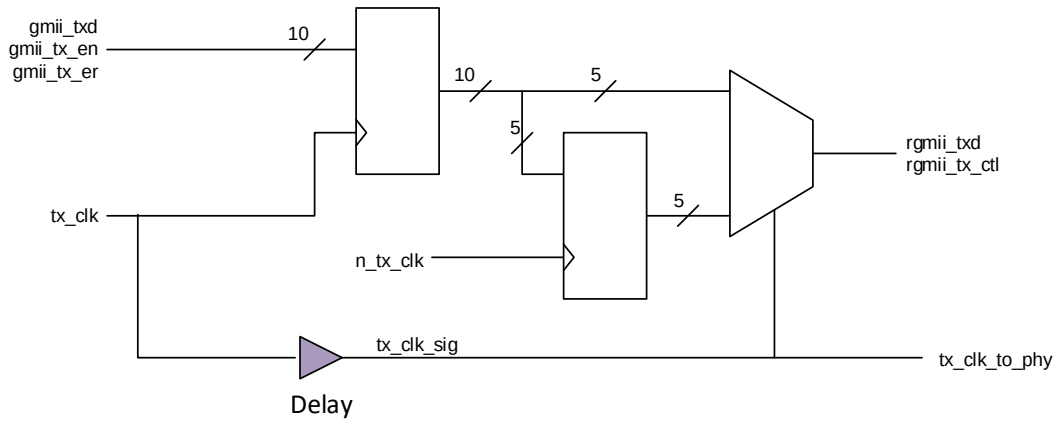
In the diagram the `gmii_txd[7:0]` data signals are sampled on the rising edge of `tx_clk` (to produce `gmii_txd_r`) and bits [7:4] of this then resampled on the rising edge of `n_tx_clk` (which is the inverted `tx_clk`). This is illustrated in the following timing diagram:



The lower nibble of the data byte `rgmii_txd[3:0]` is available on the rising edge of `tx_clk` and the upper nibble is available on the following falling edge of `tx_clk`. The signal `tx_clk_sig` selects the lower nibble to be output on `rgmii_txd` when high and the upper nibble when low.

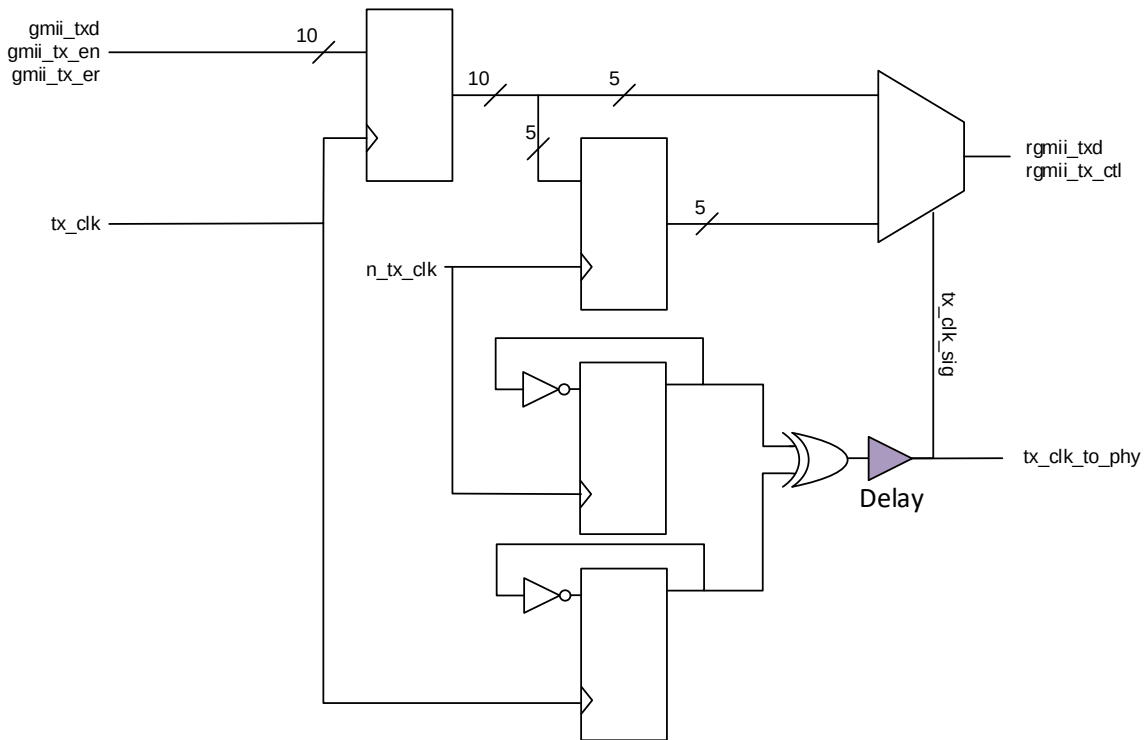
`tx_clk_sig` is an input to the GEM_GXL used to control the multiplexer logic feeding the DDR outputs. It is the responsibility of the system integrator to produce this signal, but it can be sourced from `tx_clk`. Since the main RGMII outputs directly feed chip level pads, it is generally recommended to avoid glitches on these signals. In the diagram above, `tx_clk_sig` is delayed by approximately 2ns (a 90 degree shift of `tx_clk`). It does not necessarily have to be set to this value, but delaying the mux select by a sufficient amount ensures the relevant inputs to that mux are stable when `tx_clk_sig` changes. Note that ensuring the mux outputs are glitch-free is not an RGMII specific timing requirement but is considered a sensible approach. Two examples that the integrator may choose for generating `tx_clk_sig` are shown below:

Example 1:



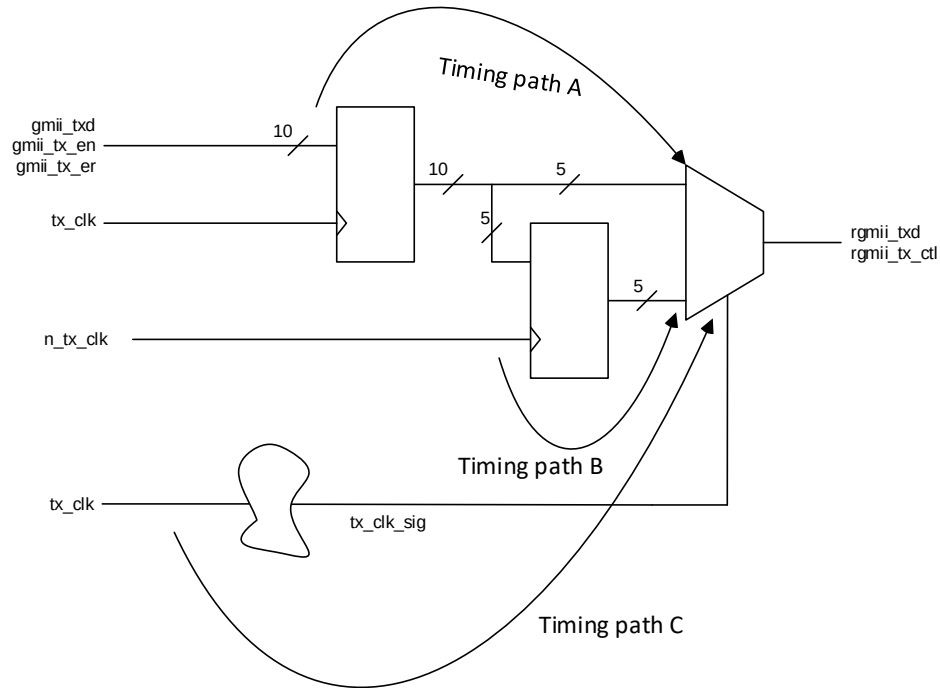
The delay here would have to be greater than the clock insertion delay to the flops plus the clock to Q delay of the flops themselves. For example, a value of 1ns should in most cases be sufficient to ensure clean transitions.

Example 2:



This example shows an implementation where `tx_clk_sig` can be generated using two toggling flops. The added delay in the diagram only needs to be marginal. The reset states of each flop and the initialization of the clocks feeding the flops must carefully be controlled to ensure the polarity of `tx_clk_sig` is correct and that it matches `tx_clk` rather than `n_tx_clk`.

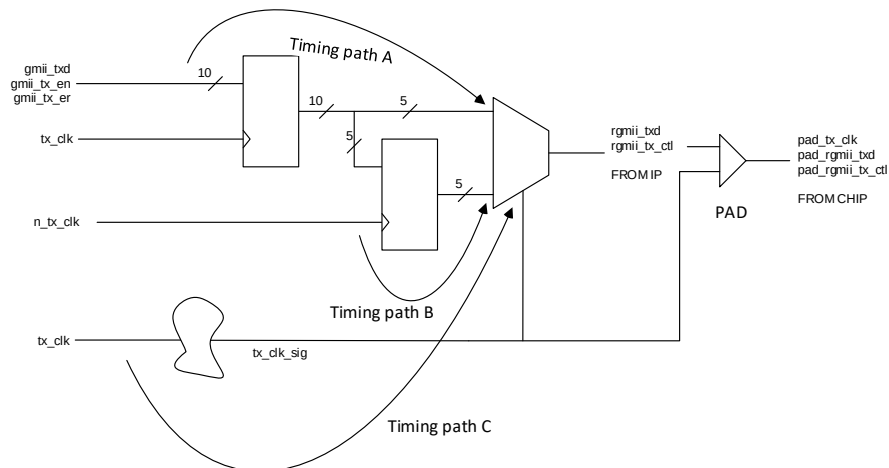
In all cases, it is recommended the integrator reviews the timing reports for the 3 key paths as shown below.



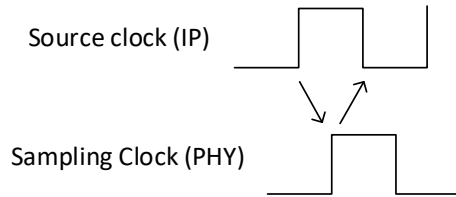
The delay shown in timing path C should be greater than both timing path A and B. STA of the 3 paths needs to include the clock tree delays and should be performed across all corners.

In terms of synthesis constraints, `rgmii_txd` and `rgmii_tx_ctl` are top level outputs and the paths from final flop to output should be tightly constrained to avoid major routing or cell delays being added. There is some margin allowed within the standard (up to 1ns drift at the receiving end) but it is recommended that the path from the `tx_clk` and `n_tx_clk` sources through the flop and mux to the outputs be constrained tightly. The RGMII standard states the variation should be within 500ps at the transmitting end before any delay to offset the clock/data is added. Similarly, the path from `tx_clk_sig` through to the outputs should be constrained similarly tightly.

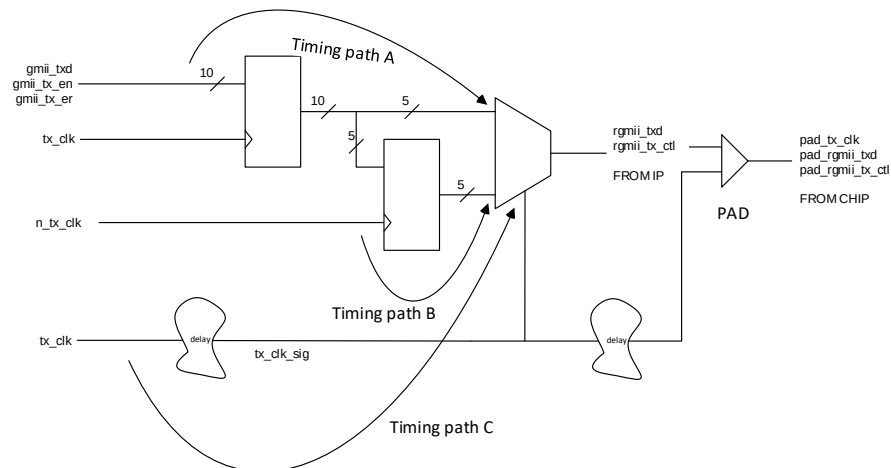
The following diagram includes the expected chip level pad, assuming no added delay to `tx_clk` output:



`tx_clk`, `rgmii_txd` and `rgmii_tx_ctl` will all output the chip via pads and will be routed across a board to the PHY. Most RGMII PHYs have a programmable delay feature built in to allow various degrees of clock shifting to help ensure stable sampling, but essentially the sampling clock within the PHY must have a relationship as follows:



This delay on the sampling clock will allow the PHY to sample the DDR data in a stable window. As previously mentioned, this delay will be either added by the RGMII PHY (if it has that option), or will be added at source within the SoC (as per RGMII-ID standard), or will be added in the PCB (original RGMII v1.3). If the decision is to implement the delay within the SoC, the diagram above changes to:



Note that from release 1p09 onwards, when RGMII is configured the core supports operating in MII mode by selecting “`sel_mii_on_rgmii`”, bit 28 of the network control register, offset 0x0. When this is active `gmii_txd` in the Transmit RGMII diagram above is replaced with `{txd[3:0], txd[3:0]}` to make the upper and lower nibbles identical. `rgmii_txd` becomes the MII `txd` output, `rgmii_tx_ctl` becomes the MII `tx_en` and a delayed version of `gmii_tx_er` is output at the top-level to create the MII `tx_er` output. `tx_clk` will be sourced from the PHY in MII mode, but the clock trees will be identical and no specific synthesis constraints considerations are needed to support this mode.